n-body (n-site) terms in ALPS, step by step

ALPS models had two kinds of terms: SITETERM (one site) and BONDTERM (two sites). This branch adds a third, NSITETERM: one operator expression on an ordered group of any number of sites, attached to a new <NSITE> element of the lattice. A molecular Hamiltonian's (pq|rs), a ring exchange or a correlated pair hop can now be written down exactly, with nothing truncated. The terms are used by exact diagonalization (`fulldiag`, `sparsediag`) and by `dmrg`.

This document follows one small problem through ALPS: the input files, the n-site group's local table, the fill of the many-body matrix, and the eigenvalues. It is the same system as quartic_ED_walkthrough.pdf (one 2×2 plaquette, t = 1, U = 4, K = 0.5, one ↑ and one ↓ electron). Every matrix and number below was printed by ALPS itself (`pdf_source/`).

Optional: nothing changes unless the input asks for it

There is no switch to turn on and nothing to change in existing inputs. The new code runs only when the lattice contains an <NSITE> and the model contains an <NSITETERM>:

lattice has <NSITE>	model has <NSITETERM>	what happens	checked
no	no	exactly as before	stock models: master vs branch bit-identical; full ctest 931/931
yes	no	n-site groups are ignored	stock 'fermion Hubbard' on this lattice: E0 = -3.4185507189
no	yes	term has nothing to act on (like a BONDTERM on a graph without edges)	K = 0.5 model, n-site group removed: E0 = -3.4185507189 (= the K = 0 value)
yes	yes	fulldiag / sparsediag / dmrg add the term; the QMC codes stop with a message	this document; dmrg on the same input: E0 = -3.3597011605 (Step 8 f)

The QMC codes refuse such a model with a message, so it is never silently truncated.

Step 1 — Lattice: an <NSITE> is an ordered group of sites

A <GRAPH> gets n-site group elements next to its edges. Each <SITE id> is a vertex id (1-based, like EDGE source/target). The order of the sites is what the model's role names bind to: here i, j, k, l = vertices 1, 2, 4, 3, i.e. ALPS sites 0, 1, 3, 2 going round the square. The n-site group is stored in its own list beside the graph; it adds no bonds.

```
<LATTICES>
  <GRAPH name="2x2 plaquette" vertices="4">
    <!--  3 ---- 4
          |      |
          1 ---- 2      (ALPS numbers sites 0..3 internally) -->
    <EDGE source="1" target="2"/>
    <EDGE source="3" target="4"/>
    <EDGE source="1" target="3"/>
    <EDGE source="2" target="4"/>
    <!-- new: an ordered group of sites; the order fixes the roles i j k l -->
    <NSITE type="0">
      <SITE id="1"/><SITE id="2"/><SITE id="4"/><SITE id="3"/>
    </NSITE>
  </GRAPH>
</LATTICES>
```

```
ALPS builds:

  sites      0 1 2 3
  bonds      (0,1) (2,3) (0,2) (1,3)
  n-site groups type 0: (0, 1, 3, 2)

     2 ----- 3
     |   K   |
     0 ----- 1

  roles  i=0  j=1  k=3  l=2
```

For a periodic lattice the n-site group goes in the <UNITCELL>, with each site given as vertex + cell offset exactly like an EDGE's SOURCE/TARGET. ALPS repeats it over the lattice, wraps it at periodic boundaries and drops it where a site would fall off an open boundary:

```
<UNITCELL name="ladder cell" dimension="1" vertices="2">
  ...edges as usual...
  <NSITE type="0">
    <SITE vertex="1" offset="0"/> <SITE vertex="2" offset="0"/>
    <SITE vertex="2" offset="1"/> <SITE vertex="1" offset="1"/>
  </NSITE>
</UNITCELL>
```

Step 2 — Model: an <NSITETERM> is written like a BONDTERM

`sites="i j k l"` names the roles; the term is any expression in the site operators (and library operators), on as many roles as the n-site group has sites. As for bond terms, `type="…"` restricts a term to one n-site group type, and without it `#` in parameter names is replaced by the type (K# → K0, K1, …). Fermion signs follow the operator order as written. The SITEBASIS and BASIS at the top of model.xml are the usual fermion ones; only the HAMILTONIAN is shown.

```
<HAMILTONIAN name="Hubbard with pair hopping">
  <PARAMETER name="t" default="1"/>
  <PARAMETER name="U" default="0"/>
  <PARAMETER name="K" default="0"/>
  <BASIS ref="fermion"/>
  <SITETERM site="i">U*n_up(i)*n_down(i)</SITETERM>
  <BONDTERM source="i" target="j">
    -t*(cdag_up(i)*c_up(j) + cdag_up(j)*c_up(i) + cdag_down(i)*c_down(j) + cdag_down(j)*c_down(i))
  </BONDTERM>
  <!-- new: one term on four sites -->
  <NSITETERM sites="i j k l">
    K*(cdag_up(l)*cdag_down(k)*c_down(j)*c_up(i) + cdag_up(i)*cdag_down(j)*c_down(k)*c_up(l))
  </NSITETERM>
</HAMILTONIAN>
```

The K term is quartic on four different sites: it moves an ↑ electron from i to l and a ↓ electron from j to k together (and back). No site or bond term can produce it.

Step 3 — Run it like any other ALPS model

```
LATTICE_LIBRARY = "lattice.xml"
MODEL_LIBRARY   = "model.xml"
GRAPH           = "2x2 plaquette"
MODEL           = "Hubbard with pair hopping"
CONSERVED_QUANTUMNUMBERS = "Nup,Ndown"
Nup_total   = 1
Ndown_total = 1
t = 1
U = 4
{ K = 0 }
{ K = 0.5 }
```

```
$ parameter2xml parm
$ fulldiag parm.in.xml

lowest levels (fulldiag, N↑ = N↓ = 1):
  K = 0    -3.418551  -2.000000  -2.000000  -0.828427
  K = 0.5  -3.359701  -2.000000  -1.882634  -1.106328
```

All 16 levels for K = 0.5: -3.359701 -2.000000 -1.882634 -1.106328 -0.828427 -0.188513 -0.000000 +0.000000 +0.123791 +1.630788 +2.000000 +2.144396 +4.000000 +4.794841 +4.828427 +5.843361. These are the numbers in quartic_ED_walkthrough.pdf (E0 = −3.3597011665, and −3.4185507189 at K = 0).

Step 4 — The ALPS basis of the (N↑ = 1, N↓ = 1) sector

Each site has 4 local states, numbered by ALPS from the quantum numbers (Nup, Ndown): 0 = · (0, 0), 1 = ↓ (0, 1), 2 = ↑ (1, 0), 3 = ↕ (1, 1) (· empty, ↕ doubly occupied). A basis state is the tuple of local states on sites 0…3, and ALPS sorts the 16 tuples that satisfy the constraints. The Jordan–Wigner order is site 0, 1, 2, 3 and ↑ before ↓ on a site, i.e. mode = 2·site + spin as in the earlier PDFs; the bit column is that occupation string.

ALPS index	local states (site 0 1 2 3)	bits m7…m0	electrons	picture (2 3 / 0 1)
0	0 0 0 3	11000000	|↑3 ↓3>	· ↕ / · ·
1	0 0 1 2	01100000	|↓2 ↑3>	↓ ↑ / · ·
2	0 0 2 1	10010000	|↑2 ↓3>	↑ ↓ / · ·
3	0 0 3 0	00110000	|↑2 ↓2>	↕ · / · ·
4	0 1 0 2	01001000	|↓1 ↑3>	· ↑ / · ↓
5	0 1 2 0	00011000	|↓1 ↑2>	↑ · / · ↓
6	0 2 0 1	10000100	|↑1 ↓3>	· ↓ / · ↑
7	0 2 1 0	00100100	|↑1 ↓2>	↓ · / · ↑
8	0 3 0 0	00001100	|↑1 ↓1>	· · / · ↕
9	1 0 0 2	01000010	|↓0 ↑3>	· ↑ / ↓ ·
10	1 0 2 0	00010010	|↓0 ↑2>	↑ · / ↓ ·
11	1 2 0 0	00000110	|↓0 ↑1>	· · / ↓ ↑
12	2 0 0 1	10000001	|↑0 ↓3>	· ↓ / ↑ ·
13	2 0 1 0	00100001	|↑0 ↓2>	↓ · / ↑ ·
14	2 1 0 0	00001001	|↑0 ↓1>	· · / ↑ ↓
15	3 0 0 0	00000011	|↑0 ↓0>	· · / ↕ ·

Orange: the two states the K term connects.

Step 5 — The n-site group's local table (NSiteOperator::matrix)

ALPS first turns the term string into a table on the 4 n-site group sites alone, indexed by tuples of local states in role order (i, j, k, l): $4^4 = 256$ input tuples. Each product in the term is split once into a number times one operator string per role, $O_i\, O_j\, O_k\, O_l$: the site operators are read right to left, and each odd (fermionic) operator moved into role order gives one sign for each odd operator it passes. Each O_r becomes a 4×4 single-site matrix. A row of the table is the product of these, assembled the first time the fill (Step 6) asks for it, so only rows that basis states reach are built. Of the 256 × 256 entries only 32 are nonzero. Signs between the n-site group and the rest of the lattice are not in the table; they are added in Step 6, as for bond terms. The table depends only on the n-site group type and its site types, so it is built once and reused on every n-site group of the lattice.

Below: the 16 entries for the first half, $K\, c\dagger_{l\uparrow}\, c\dagger_{k\downarrow}\, c_{j\downarrow}\, c_{i\uparrow}$, in units of K. The other 16 (the Hermitian conjugate) are the same with input and output swapped. There are 16 because each site may also hold the other spin as a spectator (2^4). The sign depends only on those spectators: $c_\downarrow$ acting on ↕ passes the ↑ on the same site (↑ comes first), which gives −1, while $c_\uparrow$ on ↕ gives +1.

in (i j k l)		out (i j k l)		value / K	spectators
2100	↑ ↓ · ·	0012	· · ↓ ↑	+1	none
2101	↑ ↓ · ↓	0013	· · ↓ ↕	+1	l
2120	↑ ↓ ↑ ·	0032	· · ↕ ↑	-1	k
2121	↑ ↓ ↑ ↓	0033	· · ↕ ↕	-1	k, l
2300	↑ ↕ · ·	0212	· ↑ ↓ ↑	-1	j
2301	↑ ↕ · ↓	0213	· ↑ ↓ ↕	-1	j, l
2320	↑ ↕ ↑ ·	0232	· ↑ ↕ ↑	+1	j, k
2321	↑ ↕ ↑ ↓	0233	· ↑ ↕ ↕	+1	j, k, l
3100	↕ ↓ · ·	1012	↓ · ↓ ↑	+1	i
3101	↕ ↓ · ↓	1013	↓ · ↓ ↕	+1	i, l
3120	↕ ↓ ↑ ·	1032	↓ · ↕ ↑	-1	i, k
3121	↕ ↓ ↑ ↓	1033	↓ · ↕ ↕	-1	i, k, l
3300	↕ ↕ · ·	1212	↓ ↑ ↓ ↑	-1	i, j
3301	↕ ↕ · ↓	1213	↓ ↑ ↓ ↕	-1	i, j, l
3320	↕ ↕ ↑ ·	1232	↓ ↑ ↕ ↑	+1	i, j, k
3321	↕ ↕ ↑ ↓	1233	↓ ↑ ↕ ↕	+1	i, j, k, l

Step 6 — Fill the many-body matrix (hamiltonian_matrix::apply_operator)

For every basis state, ALPS reads the local states on the n-site group's sites (in role order), looks up that row of the table, and writes each output tuple back into the state to get the target basis state. The only extra ingredient is the Jordan–Wigner sign of each odd operator O_r: $(-1)^n$, where n is the number of fermions on sites before site(r), counted in the state as already changed by the operators applied before it. For two sites this is exactly the rule ALPS already uses for bond terms. One column, worked through:

```
column 14 = |↑0 ↓1>  (picture · · / ↑ ↓),  n-site group tuple (i j k l) = 2 1 0 0
table entry -> output tuple 0 0 1 2, value +1 K
operators in role order, applied last role first:
  role l = site 2: · -> ↑   fermions on sites < 2: 2  -> even, +1
  role k = site 3: · -> ↓   fermions on sites < 3: 2 (+1: role l already applied on site 2)  -> odd, -1
  role j = site 1: ↓ -> ·   fermions on sites < 1: 1  -> odd, -1
  role i = site 0: ↑ -> ·   fermions on sites < 0: 0  -> even, +1
result: sign +1, row 2 = |↑2 ↓3>,  H[2,14] += +0.5
```

This is the entry the earlier PDF finds as V[4,7|0,3] → H[11,1] = +K in its own state numbering. Re-implementing this rule in numpy (make_pdf.py, `fill`) on the 32 table entries reproduces ALPS's K contribution exactly: max |difference| = 0e+00.

Step 7 — The full 16 × 16 Hamiltonian, as ALPS builds it

Rows and columns use the ALPS basis of Step 4. Yellow: U on doubly occupied sites (site terms). Orange: the K term (n-site term). Everything else: hopping (bond terms), whose ± sign comes from the same Jordan–Wigner rule.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	+4	+1	-1	·	+1	·	-1	·	·	·	·	·	·	·	·	·
1	+1	·	·	+1	·	·	·	+1	·	-1	·	·	·	·	·	·
2	-1	·	·	-1	·	+1	·	·	·	·	·	·	-1	·	+0.5	·
3	·	+1	-1	+4	·	·	·	·	·	·	+1	·	·	-1	·	·
4	+1	·	·	·	·	-1	·	·	+1	-1	·	·	·	·	·	·
5	·	·	+1	·	-1	·	·	·	·	·	-1	·	·	·	+1	·
6	-1	·	·	·	·	·	·	-1	-1	·	·	·	-1	·	·	·
7	·	+1	·	·	·	·	-1	·	·	·	·	+1	·	-1	·	·
8	·	·	·	·	+1	·	-1	·	+4	·	·	+1	·	·	-1	·
9	·	-1	·	·	-1	·	·	·	·	·	-1	-1	·	·	·	·
10	·	·	·	+1	·	-1	·	·	·	-1	·	·	·	·	·	+1
11	·	·	·	·	·	·	·	+1	+1	-1	·	·	·	·	·	+1
12	·	·	-1	·	·	·	-1	·	·	·	·	·	·	-1	-1	·
13	·	·	·	-1	·	·	·	-1	·	·	·	·	-1	·	·	-1
14	·	·	+0.5	·	·	+1	·	·	-1	·	·	·	-1	·	·	-1
15	·	·	·	·	·	·	·	·	·	·	+1	+1	·	-1	-1	+4

Check against the independent prototype. Re-indexed into the earlier PDF's state order, ALPS's matrix equals the prototype's (hyper_ed.py: normal-ordered coefficient tables and bit-string fill, no ALPS code) entry by entry, signs included: max |difference| = 0.0. Eigenvalues: E0 = -3.3597011665 (K = 0.5), -3.4185507189 (K = 0).

Step 8 — Benchmarks

(a) Molecules: ALPS ED vs QDK CASCI, full (pq|rs). A molecular Hamiltonian's two-body part touches 3 and 4 distinct orbitals, so before this change it could not be written for ALPS without truncation. Now every orbital set touched by a nonzero (pq|rs) becomes an <NSITE> with its own <NSITETERM>. All orbitals active, STO-3G. The last column compares the whole spectrum with a dense Jordan–Wigner diagonalization of the same integrals (numpy, independent of both codes). Result: ALL AGREE.

molecule	sector (N↑,N↓)	n-site groups 2/3/4-site	QDK CASCI	ALPS ED	|ΔE|	spectrum vs dense JW
H4 chain 0.9 A	(2,2)	6/4/1	-2.1803166186	-2.1803166186	1.8e-15	7.1e-15
H4 chain 0.9 A	(2,1)	6/4/1	-1.7659014705	-1.7659014705	4.4e-15	3.6e-15
H4 chain 0.9 A	(3,1)	6/4/1	-1.8916100906	-1.8916100906	2.7e-15	2.7e-15
H4 rectangle 0.8x1.2 A	(2,2)	6/0/1	-2.0922392274	-2.0922392274	8.9e-16	9.3e-15
H4 rectangle 0.8x1.2 A	(3,1)	6/0/1	-1.8188072640	-1.8188072640	3.6e-15	6.2e-15
H4 bent/asym	(2,2)	6/4/1	-2.1573698719	-2.1573698719	7.1e-15	5.3e-15
H4 bent/asym	(2,1)	6/4/1	-1.7521256324	-1.7521256324	4.4e-15	4.0e-15
H4 bent/asym	(1,1)	6/4/1	-0.9132368580	-0.9132368580	1.8e-15	4.0e-15
H3+ triangle	(1,1)	3/1/0	-1.2693966626	-1.2693966626	4.4e-16	4.0e-15
H3+ triangle	(2,0)	3/1/0	-0.6869268798	-0.6869268798	0.0e+00	8.9e-16
H4 chain 1.5 A (stretched)	(2,2)	6/4/1	-1.9961503613	-1.9961503613	2.7e-15	4.9e-15
H4 chain 1.5 A (stretched)	(3,1)	6/4/1	-1.9255585303	-1.9255585303	1.8e-15	4.4e-15
LiH 1.6 A (6 orbitals)	(2,2)	15/16/1	-7.8823243948	-7.8823243948	5.3e-15	3.9e-14
LiH 1.6 A (6 orbitals)	(2,1)	15/16/1	-7.6141564878	-7.6141564878	7.1e-15	1.6e-14
H6 ring 1.0 A	(3,3)	15/14/9	-3.2374767382	-3.2374767382	1.8e-15	4.1e-14

(b) Exact answer on 4 sites. Open 4-site chain plus U on one delocalized hopping eigenmode $a = \Sigma\, \varphi_i\, c_i$: $H = -t\, \Sigma$ hop $+ U\, n_{a\uparrow}\, n_{a\downarrow}$. Written in the site basis, the U term is one NSITETERM with 1-, 2-, 3- and 4-site pieces. The spectrum is known exactly: $\Sigma\, \varepsilon_{\text{occupied}} + U\cdot$[mode a doubly occupied]. Checked for two modes, three role orderings and five sectors. A term that changes N by 4 is checked in the full 256-state space against a dense matrix, and with N↑, N↓ conserved it is refused. 32 runs; ALL OK.

(c) Ladders vs the prototype. 2×L Hubbard ladders with a pair-hopping n-site term, open and periodic, several sectors: ALL OK.

(d) Random two-body terms on 2- and 3-site groups, all index patterns, with interleaved site orders, vs a dense Jordan–Wigner matrix: ALL OK.

(e) Bosonic Moore–Read state: a 3-body term, 6 operators. Bosons in the lowest Landau level on a sphere with 2S flux; the orbitals are the sites. The 3-body contact interaction $H = \Sigma_M T_M^\dagger T_M$ projects every triple onto total angular momentum 3S, and is written as one NSITETERM over all orbitals (strings like $b^\dagger_0 b^\dagger_0 b^\dagger_2 b_1 b_1 b_1$). The spectrum is compared with a dense reference; the number of zero modes with the number of patterns with at most 2 bosons in any 2 neighbouring orbitals (Bernevig–Haldane). At 2S = N − 2 the zero mode is unique: the Moore–Read ground state. ALL OK.

N	2S	dim	max|ΔE| vs dense	zero modes	admissible	gap
4	2	15	6.22e-15	1	1	7.200000
4	3	35	1.07e-14	6	6	5.785714
4	4	70	1.42e-14	20	20	5.672727
6	4	210	1.18e-13	1	1	5.703896
6	5	462	1.28e-13	10	10	4.846328
6	6	924	1.85e-13	50	50	4.606700
8	6	3003	1.82e-12	1	1	5.025037

(f) DMRG. `dmrg` takes the same input. Each product of the term becomes one DMRG term with one site operator per site (the split of Step 5, before the table is built); operators in the same block are combined into block operators, as for bond terms between distant sites. On this document's example it gives E0 = -3.3597011605 (fulldiag -3.3597011665). Molecules, every (pq|rs) an n-site term on 2–4 orbitals, against QDK CASCI (H3+ is left out: dmrg needs at least 4 sites, for any model): ALL AGREE.

molecule	sector (N↑,N↓)	QDK CASCI	ALPS dmrg	|ΔE|
H4 chain 0.9 A	(2,2)	-2.1803166186	-2.1803166186	2.2e-13
H4 chain 0.9 A	(2,1)	-1.7659014705	-1.7659014705	1.1e-14
H4 chain 0.9 A	(3,1)	-1.8916100906	-1.8916100906	8.9e-16
H4 rectangle 0.8x1.2 A	(2,2)	-2.0922392274	-2.0922392274	6.7e-14
H4 rectangle 0.8x1.2 A	(3,1)	-1.8188072640	-1.8188072640	5.3e-15
H4 bent/asym	(2,2)	-2.1573698719	-2.1573698719	5.6e-14
H4 bent/asym	(2,1)	-1.7521256324	-1.7521256324	1.8e-15
H4 bent/asym	(1,1)	-0.9132368580	-0.9132368580	2.4e-11
H4 chain 1.5 A (stretched)	(2,2)	-1.9961503613	-1.9961503613	9.8e-15
H4 chain 1.5 A (stretched)	(3,1)	-1.9255585303	-1.9255585303	4.0e-15
LiH 1.6 A (6 orbitals)	(2,2)	-7.8823243948	-7.8823243948	1.8e-14
LiH 1.6 A (6 orbitals)	(2,1)	-7.6141564878	-7.6141564878	3.6e-15
H6 ring 1.0 A	(3,3)	-3.2374767382	-3.2374767382	1.6e-14

Lattice cases (benchmarks/alps_checks/t7_dmrg.py): ladders vs fulldiag; random 4-operator and 6-operator (3-body) fermion terms on 4- and 3-site n-site groups in scrambled site order vs a dense Jordan–Wigner matrix (E0 and E1, odd sectors included); Moore–Read; a 2×6 ladder where DMRG truncates, vs sparsediag; the conserved-quantum-number refusal and the Nmax warning. ALL OK.

With `SAVE_MPS=1` (ALPSim/ALPS#216) dmrg also writes its ground state as an MPS. Contracted to a state vector, it matches the CASCI wave function of every molecule above determinant by determinant (ALL AGREE), and the exact ground states of fermion, boson and spin models with n-site terms component by component (ALL OK): benchmarks/results/casci_vs_mps.txt, t8_mps.txt.

```
  2x2 demo K=0.5 (1,1)                        max|dE| 5.8e-15  E0 -3.3597011665
  2x3 open U=8 K=0.7 (2,2)                    max|dE| 5.3e-15  E0 -5.3669606044
  2x3 open U=8 K=0.7 (3,2)                    max|dE| 2.2e-13  E0 -3.4189803903
  2x3 periodic U=8 K=0.9 (3,2)                max|dE| 1.9e-14  E0 -5.0117188128
  2x4 open U=8 K=1.1 (3,2)                    max|dE| 4.7e-14  E0 -7.8586086456
  2x4 open U=8 K=1.1 (3,3)                    max|dE| 3.9e-13  E0 -5.6019556305
  6 sites, sector (1,1)                       max|dE| 1.8e-15  E0 -6.3249583371
  6 sites, sector (2,1)                       max|dE| 1.8e-14  E0 -9.3268308972
  6 sites, sector (3,2)                       max|dE| 2.3e-14  E0 -15.9045105124
  6 sites, sector (3,3)                       max|dE| 2.1e-14  E0 -17.4538532381
  6 sites, sector (4,2)                       max|dE| 2.5e-14  E0 -16.7924619556
  N=6 2S=4 (Moore-Read ground state)           max|dE| 1.4e-13  E0 -5.8e-17 gap 5.703896
  N=8 2S=6 (Moore-Read ground state)           max|dE| 2.1e-12  E0 1.6e-15 gap 5.025037
  2x6 open U=8 K=1.1 (4,4), 400 states        max|dE| 3.2e-09  E0 -13.3426010596  sparsediag 10s dmrg 8s
5. refusals and warnings
  N-changing term with N conserved is refused  yes
  Nmax cutoff warning in fulldiag              yes
  Nmax cutoff warning in dmrg                  yes
```

(g) Unit tests. src/alps/model/tests/nsite_terms.cpp (GoogleTest): unit-cell n-site groups at open and periodic boundaries, GRAPH n-site groups, invalid n-site groups, ladder reference energies, K = 0 equals the plain model, the analytic case above, sector violation, typed and # terms, parameter precision, the refusal in other applications, and bosons: N-conserving and non-conserving terms vs a dense reference, truncation at Nmax and its warning. For dmrg, src/apps/dmrg/dmrg/tests/dmrg_nsite_terms.cmake runs two ladders with the pair-hopping term against their fulldiag energies. Full ctest: 937/937 pass.

Raw output and the scripts are in `benchmarks/`.

Step 9 — What changed in ALPS

file	change
lattice/nsite_group.h (new)	n-site group {type, sites}; reading <NSITE> in GRAPH and UNITCELL
lattice/graph.h, latticelibrary.h/.C	GRAPH accepts <NSITE>; the library keeps them per graph
lattice/unitcell.h/.C	UNITCELL accepts <NSITE> (vertex + offset), and writes it back
lattice/latticegraph.h	unit-cell n-site groups repeated over the lattice, same boundary rules as edges
lattice/graph_helper.h	nsite_groups(), num_nsite_groups()
model/nsiteoperator.h (new)	NSiteOperator, the splitter into per-site operators with the sign, the local table
model/nsiteterm.C (new)	read/write <NSITETERM>
model/globaloperator.h/.C, default_term.h	HAMILTONIAN holds n-site terms (typed and #)
model/hamiltonian_matrix.hpp	n-site group loop: local table per type, Jordan–Wigner fill, sector check
model/model_helper.h	applications without support refuse n-site terms instead of dropping them
model/siteterm.C, bondterm.C	fix: parameters were rounded to 6 digits (see below)
model/tests/nsite_terms.cpp (new)	unit tests
apps/dmrg/dmrg/dmrg.h	n-site terms as DMRG terms, one site operator per site; same checks as ED
apps/dmrg/dmrg/tests/ dmrg_nsite_terms.cmake (new)	dmrg test against fulldiag energies
CHANGELOG.md, model/docs/	changelog entry, syntax reference (nsite_terms.md), this folder

One fix to existing behaviour. SITETERM and BONDTERM substituted parameter values into the expression with 6 significant digits, so `t0=0.123456789` acted as 0.123457. Molecular integrals exposed this as a ~$5 \cdot 10^{-6}$ error against CASCI. Values are now written with full double precision. Inputs with short parameter values (t = 1, U = 4, …) give bit-identical results; longer values are now used exactly as given.

Limits. The QMC codes refuse n-site terms. In dmrg every distinct product is its own term with its own block operators, so the cost grows with the number of products ($O(N^4)$ for a molecule with N orbitals). N-site groups in a <GRAPH> are not yet written back out when ALPS writes a lattice library to XML.

Appendix — the fill (Step 6) in ALPS

src/alps/model/include/alps/model/hamiltonian_matrix.hpp:

```
// Fill one n-site term on n-site group p. The local table (see nsiteoperator.h)
// stores each product reordered as O_0 O_1 ... O_{n-1} (role order). Acting on
// a basis state, O_{n-1} is applied first; each odd O_r picks up the
// Jordan-Wigner sign (-1)^(number of fermions on sites before p.sites[r]),
// counted in the state as modified by the operators already applied. This is
// the rule the bond fill below encodes in closed form for n = 2.
template <class M, class G> template <class STATES, class V, class W>
void hamiltonian_matrix<M,G>::apply_operator(const STATES& states, const nsite_matrix<value_type>& mat,
      const nsite_group& p, const V& x, W& y) const
{
  if (!built_basis_)                    // the parity table below reads basis_
    build_basis();
  const std::size_t n = p.sites.size();
  const std::size_t nsites = graph_.num_sites();
  // fermion parity of every local state of every site
  std::vector<std::vector<bool> > parity(nsites);
  for (std::size_t s=0;s<nsites;++s)
    for (std::size_t k=0;k<basis_[s].size();++k)
      parity[s].push_back(is_fermionic(model_.model().basis().site_basis(graph_.site_type(s)),basis_[s][k]));

  std::vector<std::size_t> in(n), out(n);
  std::vector<bool> prefix(nsites+1);
  for (std::size_t i=0;i<dimension();++i) {         // loop over source states
    state_type state=states[i];
    for (std::size_t r=0;r<n;++r)
      in[r]=state[p.sites[r]];
    const typename nsite_matrix<value_type>::row_type& row = mat.row(mat.index(in));
    if (row.empty())
      continue;
    prefix[0]=false;                                // parity of sites < s
    for (std::size_t s=0;s<nsites;++s)
      prefix[s+1] = prefix[s] != parity[s][state[s]];
    for (std::size_t e=0;e<row.size();++e) {
      mat.decode(row[e].first,out);
      state_type newstate=state;
      for (std::size_t r=0;r<n;++r)
        newstate[p.sites[r]]=out[r];
      std::complex<double> phase;
      std::size_t j;
      boost::tie(j,phase) = states.index_and_phase(newstate);
      if (j>=dimension()) {
        if (building_hamiltonian_)
          check_sector(state,newstate,"<NSITETERM> of type "+boost::lexical_cast<std::string>(p.type),p.sites);
        continue;
      }
      bool f=false;
      for (std::size_t r=n;r-->0;) {
        std::size_t s=p.sites[r];
        if (parity[s][in[r]]!=parity[s][out[r]]) {  // O_r is odd
          bool before=prefix[s];
          for (std::size_t q=r+1;q<n;++q)           // already applied O_q
            if (p.sites[q]<s && parity[p.sites[q]][in[q]]!=parity[p.sites[q]][out[q]])
              before=!before;
          if (before)
            f=!f;
        }
      }
      value_type val=row[e].second;
      if (f)
        val=-val;
      val *= numeric_cast<value_type>(phase * states.normalization(j)/states.normalization(i));
      y[i] += val*x[j];                  // set matrix element
      simplify(y[i]);               // optionally simplify a symbolic expression
    }
  }
}
```

The sign used when building the local table (Step 5), src/alps/model/include/alps/model/nsiteoperator.h:

```
// Operators are applied right to left (OperatorEvaluator::direction), so all
// operators already seen stand to the right of the current one. Moving the
// current operator on role r into role order means moving it right past the
// already-seen operators on roles q < r: one sign per odd one among them.
// For n = 2 this is BondOperatorSplitter's rule with the roles swapped.
template <class I, class T>
expression::Expression<T> NSiteOperatorSplitter<I, T>::partial_evaluate_function(const std::string& name,
      const expression::Expression<T>& arg, bool isarg) const
{
  int r = role(arg);
  if (r >= 0) {
    // Replaced by its sign, an operator inside another function such as
    // sqrt(n(i)) would enter the table as n(i) itself.
    if (isarg)
      boost::throw_exception(std::runtime_error("n-site term: the site operator " + name + "("
        + boost::lexical_cast<std::string>(arg) + ") is the argument of another function, which is not supported"));
```

```
    ops_[r] = expression::Function<T>(name, arg) * ops_[r];
    used_[r] = true;
    double sign = 1.;
    if (bases_[r].is_fermionic(name)) {
      for (int q = 0; q < r; ++q)
        if (odd_[q])
          sign = -sign;
      odd_[r] = !odd_[r];
    }
    return expression::Expression<T>(sign);
  }
  return expression::ParameterEvaluator<T>(*this).partial_evaluate_function(name, arg, isarg);
}
```